Guideline to make and understand Unit Test Case

1. Overview

- In the template, Unit test cases are based on functions. Each sheet presents test cases for one function.
- **Cover**: General information of the project and Unit Test cases
- **FunctionList**: The list of Classes and Functions in the document.
  + To control that the number of Unit TC meets customer's requirement or the norm, user should fill value for 'Normal number of Test cases/KLOC'.
  + Click on Function link to open the related Test cases of the function.
  Note: You should create new Function sheet before creating the link
- **Test Report**: provive the overview results of Functions Unit test: Test coverage, Test successful coverage (Summary, for normal/abnormal/boundary cases)
  Note: Should check the formula of "Sub Total" if you add more functions

2. Content in Test function sheet

2.1 Combination of test cases.

- To verify that number of Unit TC meets customer's requirement or not. User has to fill number LOC of tested function and fill value of 'Normal number test cases/KLOC' item in FunctionList sheet, which is required by
- If the number of Unit TC does not meet the requirement, creator should explain the reasons.
- If the number of 'Normal number test cases/KLOC' item in FunctionList sheet is not recorded, the number in

2.2 Condition and confirmation of Test cases.

Each test case is the combination of condition and confirmation.

a. Condition:

- Condition is combination of precondition and values of inputs.
- Precondition: it is setting condition that must exist before execution of the test case.
  Example: file A is precondition for the test case that needs to access file A.
- Values of inputs: it includes 3 types of values: normal, boundary and abnormal.
  . Normal values are values of inputs used mainly and usually to ensure the function works.
  . Boundary values are limited values that contain upper and lower values.
  . Abnormal values are non-expected values. And normally it processes exception cases.
- For examples:
  Input value belongs to 5<= input <=10.
  . 6,7,8,9 are normal values.
  . 5, 10 are boundary values.
  . -1, 11,... are abnormal values.

b. Confirmation:

- It is combination of expected result to check output of each function.
  If the results are the same with confirmation, the test case is passed, other case it is failed.
- Confirmation can include:
  + Output result of the function.
  + Output log messages in log file.
  + Output screen message...

c. Type of test cases and result:

- Type of test case: It includes normal, boundary and abnormal test cases. User selects the type based on
- Test case result: the actual output results comparing with the Confirmation.
  P for Passed and **F** for Failed cases.
  It can 'OK' or 'NG' (it depends on habit of the teams or customers)

2.3. Other items:

- Function Code: it is ID of the function and updated automatically according to FunctionList sheet.
- Function Name: it is name of the function and updated automatically according to FunctionList sheet.

- Created By: Name of creator.
- Executed By: Name of person who executes the unit test
- Lines of code: Number of Code line of the function.
- Test requirement: Brief description about requirements which are tested in this function, it is not mandatory.